40.0.1 UGC NET CSE | December 2015 | Part 3 | Question: 27

There are exactly ____ different finite automata with three states x, y and z over the alphabet $\{a, b\}$ where x is always the start state

A. 64 B. 256 C. 1024 D. 5832

ugcnetcse-dec2015-paper3 theory-of-computation

Answer key

40.0.2 UGC NET CSE | September 2013 | Part 2 | Question: 18

The number of 8-bit strings beginning with either 111 or 101 is

A. 64 B. 128 C. 265 D. None of the above

theory-of-computation ugcnetsep2013ii

Answer key

40.0.3 UGC NET CSE | December 2012 | Part 3 | Question: 57

Given the following statements:

I. Recursive enumerable sets are closed under complementation.
II. Recursive sets are closed under complementation.

Which is/are the correct statement?

A. only I B. only II C. both I and II D. neither I nor II

theory-of-computation ugcnetcse-dec2012-paper3

Answer key

40.0.4 UGC NET CSE | December 2012 | Part 3 | Question: 45

Which of the following regular expression identifies are true?

A. (r+s)*=r*s* B. (r+s)*=r*+s* C. (r+s)*=(r*s*)* D. r*s*=r*+s*

theory-of-computation ugcnetcse-dec2012-paper3

Answer key

40.0.5 UGC NET CSE | June 2012 | Part 3 | Question: 25, UGCNET-Dec2013-III: 42

Which is not the correct statement(s)?

I. Every context sensitive language is recursive
II. There is a recursive language that is not context sensitive

A. I is true II is false
B. I is true and II is true
C. I is false II is false
D. I is false and II is true

ugcnetcse-june2012-paper3 ugcnetcse-dec2013-paper3 theory-of-computation

Answer key

40.0.6 UGC NET CSE | June 2012 | Part 3 | Question: 52

Match the following :

(i)	Regular grammer	(a)	Pushdown automaton
(ii)	Context free grammer	(b)	Linear bounded automaton
(iii)	Unrestricted grammer	(c)	Deterministic finite automaton
(iv)	Context sensitive grammer	(d)	Turing machine

A. (i)-(c), (ii)-(a), (iii)-(b), (iv)-(d)
B. (i)-(c), (ii)-(a), (iii)-(b), (iv)-(d)
C. (i)-(c), (ii)-(b), (iii)-(a), (iv)-(d)
D. (i)-(c), (ii)-(b), (iii)-(d), (iv)-(a)

ugcnetcse-june2012-paper3 theory-of-computation

Answer key

40.0.7 UGC NET CSE | December 2012 | Part 2 | Question: 7

The ‘C’ language is

A. Context free language
B. Context sensitive language
C. Regular language
D. None of the above

ugcnetcse-dec2012-paper2 theory-of-computation

Answer key

40.0.8 UGC NET CSE | December 2012 | Part 3 | Question: 50

Which of the following definitions generates the same Language as L, where $L = \{WW^R \mid W \in \{a,b\}*\}$?

A. $S \to asb \mid bsa \mid \in$
B. $S \to asa \mid bsb \mid \in$
C. $S \to asa \mid bsa \mid asa \mid bsb \mid \in$
D. $S \to asa \mid bsa \mid asa \mid bsb \mid$

ugcnetcse-dec2012-paper3 theory-of-computation

Answer key

40.0.9 UGC NET CSE | June 2013 | Part 2 | Question: 24

Given a Non-deterministic Finite Automaton (NFA) with states p and r as initial and final states respectively and transition table as given below :

	a	b
p	-	q
q	r	s
r	r	s
s	r	s

The minimum number of states required in Deterministic Finite Automaton (DFA) equivalent to NFA is

A. 5
B. 4
C. 3
D. 2

ugcnetcse-june2013-paper2 theory-of-computation

Answer key

40.0.10 UGC NET CSE | June 2013 | Part 3 | Question: 36

The grammar with production rules $S \to aSb \mid SS \mid \lambda$ generates language L given by:

A. $L = \{w \in \{a,b\}* \mid n_a(w) = n_b(w)$ and $n_a(v) \geq n_b(v)$ where v is any prefix of w$\}$
B. $L = \{w \in \{a,b\}* \mid n_a(w) = n_b(w)$ and $n_a(v) \leq n_b(v)$ where v is any prefix of w$\}$
C. $L = \{w \in \{a,b\}* \mid n_a(w) \neq n_b(w)$ and $n_a(v) \geq n_b(v)$ where v is any prefix of w$\}$
D. $L = \{w \in \{a,b\}* \mid n_a(w) \neq n_b(w)$ and $n_a(v) \leq n_b(v)$ where v is any prefix of w$\}$

ugcnetcse-june2013-paper3 theory-of-computation

Answer key

40.0.11 UGC NET CSE | June 2013 | Part 3 | Question: 39

Match the following :

a.	Context sensitive language	i.	Deterministic finite automation
b.	Regular grammar	ii.	Recusive enumerable
c.	Conext free grammar	iii.	Recursive language
d.	Unrestricted grammar	iv.	Pushdown automation

Codes:

A. a-ii, b-i, c-iv, d-iii
B. a-iii, b-iv, c-i, d-ii
C. a-iii, b-i, c-iv, d-ii
D. a-ii, b-iv, c-i, d-iii

ugcnetcse-june2013-paper3 theory-of-computation match-the-following easy

Answer key

40.0.12 UGC NET CSE | June 2013 | Part 3 | Question: 40

The statements s1 and s2 are given as:

s1: Context sensitive languages are closed under intersection, concatenation, substitution and inverse homomorphism.

s2: Context sensitive languages are closed under concatenation, substitution and homomorphism.

Which of the following is correct statement?

A. Both s1 and s2 are correct
B. s1 is correct s2 is not correct
C. s1 is not correct s2 is correct
D. Both s1 and s2 are not correct

ugcnetcse-june2013-paper3 theory-of-computation

Answer key

40.0.13 UGC NET CSE | December 2013 | Part 3 | Question: 21

Given the following statements:

S_1 : The grammars $S \to asb \mid bsa \mid ss \mid a$ and $S \to asb \mid bsa \mid a$ are not equivalent.

S_2 : The grammars $S \to ss \mid sss \mid asb \mid bsa \mid \lambda$ and $S \to ss \mid asb \mid bsa \mid \lambda$ are equivalent.

A. S_1 is correct and S_2 is not correct
B. Both S_1 ad S_2 are correct
C. S_1 is not correct and S_2 is correct
D. Both S_1 ad S_2 are not correct

ugcnetcse-dec2013-paper3 theory-of-computation

Answer key

40.0.14 UGC NET CSE | December 2013 | Part 3 | Question: 41

The Greibach normal form grammar for the language $L = \{a^n b^{n+1} \mid n \geq 0\}$ is

A. $S \to aSB, B \to bB \mid \lambda$
B. $S \to aSB, B \to bB \mid b$
C. $S \to aSB \mid b, B \to b$
D. $S \to aSb \mid b$

ugcnetcse-dec2013-paper3 theory-of-computation

Answer key

40.0.15 UGC NET CSE | Junet 2015 | Part 3 | Question: 63

Given the following grammars:

G_1 $S \to AB \mid aaB$
$A \to aA \mid \epsilon$
$B \to bB \mid \epsilon$

G_2: $S \to A \mid B$

$A \rightarrow aAb \mid ab$
$B \rightarrow abB \mid \epsilon$

Which of the following is correct?

A. G_1 is ambiguous and G_2 is unambiguous grammars
B. G_1 is unambiguous and G_2 is ambiguous grammars
C. both G_1 and G_2 are ambiguous grammars
D. both G_1 and G_2 are unambiguous grammars

ugcnetcse-june2015-paper3 theory-of-computation compiler-design

Answer key

40.0.16 UGC NET CSE | December 2015 | Part 3 | Question: 73

Consider Language A defined over the alphabet $\Sigma = \{0,1\}$ as $A = \{0^{\lfloor n/2 \rfloor} \; 1^n : n >= 0\}$

The expression $\lfloor n/2 \rfloor$ means the floor of $n/2$, or what you get by rounding $n/2$ down to the nearest integer.

Which of the following is not an example of a string in A?

A. 011 B. 0111 C. 0011 D. 001111

ugcnetcse-dec2015-paper3 theory-of-computation

Answer key

40.0.17 UGC NET CSE | December 2014 | Part 3 | Question: 63

According to pumping lemma for context free languages :

Let L be an infinite context free language, then there exists some positive integer m such that any $w \in L$ with $|w| \geq m$ can be decomposed as $w = uvxyz$

A. With $|vxy| \leq m$ such that $uv^ixy^iz \in L$ for all $i = 0, 1, 2$
B. With $|vxy| \leq m$, and $|vy| \geq 1$, such that $uv^ixy^iz \in L$ for all $i = 0, 1, 2, \ldots\ldots.$
C. With $|vxy| \geq m$, and $|vy| \leq 1$, such that $uv^ixy^iz \in L$ for all $i = 0, 1, 2, \ldots\ldots.$
D. With $|vxy| \geq m$, and $|vy| \geq 1$, such that $uv^ixy^iz \in L$ for all $i = 0, 1, 2, \ldots\ldots.$

ugcnetcse-dec2014-paper3 theory-of-computation

Answer key

40.0.18 UGC NET CSE | December 2015 | Part 2 | Question: 15

In general, in a recursive and non-recursive implementation of a problem (program):

A. Both time and space complexities are better in recursive than in non-recursive program
B. Both time and space complexities are better in non-recursive than in recursive program
C. Time complexity is better in recursive version but space complexity is better in non-recursive version of a program
D. Space complexity is better in recursive version but time complexity is better in non-recursive version of a program

ugcnetcse-dec2015-paper2 theory-of-computation

Answer key

40.0.19 UGC NET CSE | December 2015 | Part 3 | Question: 23

Match the following :

	List – I		List – II
(a)	$\{a^n b^n \mid n > 0\}$ is a deterministic context free language	(i)	but not recursive language
(b)	The complement of $\{a^n b^n a^n \mid n > 0\}$ is a context free language	(ii)	but not context free language
(c)	$\{a^n b^n a^n\}$ is a context sensitive language	(iii)	but cannot be accepted by a deterministic pushdown automaton
(d)	L is a recursive language	(iv)	but not regular

Codes :

A. (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)
B. (a)-(i), (b)-(ii), (c)-(iv), (d)-(iii)
C. (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
D. (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)

ugcnetcse-dec2015-paper3 theory-of-computation match-the-following

Answer key

40.0.20 UGC NET CSE | June 2009 | Part 2 | Question: 35

Any syntactic construct that can be described by a regular expression can also be described by a :

A. Context sensitive grammar
B. Non context free grammar
C. Context free grammar
D. None of the above

ugcnetcse-june2009-paper2

Answer key

40.0.21 UGC NET CSE | July 2018 | Part 2 | Question: 35

To obtain a string of n Terminals from a given Chomsky normal from grammar, the number of productions to be used is

A. $2n-1$
B. $2n$
C. $n+1$
D. n^2

ugcnetcse-july2018-paper2 theory-of-computation conjunctive-normal-form

Answer key

40.0.22 UGC NET CSE | December 2018 | Part 2 | Question: 35

Consider the following languages:

$L_1 = \{a^{n+m} b^n a^m \mid n, m \geq 0\}$

$L_2 = \{a^{n+m} b^{n+m} a^{n+m} \mid n, m \geq 0\}$

Which of the following is correct ?

Code :

A. Only L_1 is context free language
B. Only L_2 is context free language
C. Both L_1 and L_2 are context free languages
D. Both L_1 and L_2 are not context free languages

ugcnetcse-dec2018-paper2 theory-of-computation

Answer key

40.0.23 UGC NET CSE | December 2018 | Part 2 | Question: 34

Consider the following two languages :

$L_1 = \{x \mid$ for some y with $\mid y \mid = 2^{|x|}, xy \in$ L and L is regular language$\}$

$L_2 = \{x \mid$ for some y such that $\mid x \mid = \mid y \mid, xy \in$ L and L is regular language$\}$

Which one of the following is correct ?

A. Only L_1 is regular language
B. Only L_2 is regular language
C. Both L_1 and L_2 are regular languages
D. Both L_1 and L_2 are not regular languages

ugcnetcse-dec2018-paper2 theory-of-computation

Answer key

40.0.24 UGC NET CSE | December 2004 | Part 2 | Question: 3

The context-free languages are closed for :

i. Intersection
ii. Union
iii. Complementation
iv. Kleene Star

A. (i) and (iv) B. (i) and (iii) C. (ii) and (iv) D. (ii) and (iii)

ugcnetcse-dec2004-paper2

Answer key

40.0.25 UGC NET CSE | December 2006 | Part 2 | Question: 1

Which of the regular expressions corresponds to this grammar ?
$S \to AB/AS, A \to a/aA, B \to b$

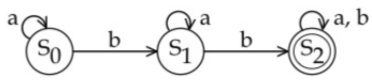
A. aa^*b^+ B. aa^*b C. $(ab)^*$ D. $a(ab)^*$

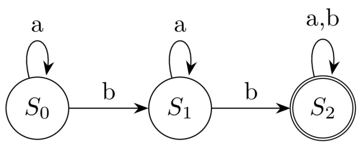
ugcnetcse-dec2006-paper2

Answer key

40.0.26 UGC NET CSE | December 2007 | Part 2 | Question: 5

Consider a Moore machine M whose digraph is :





Then $L(M)$, the language accepted by the machine M, is the set of all strings having :

A. two or more b's B. three or more b's
C. two or more a's D. three or more a's

ugcnetcse-dec2007-paper2

Answer key

40.0.27 UGC NET CSE | December 2007 | Part 2 | Question: 3

A context free grammar is :

A. type 0 B. type 1 C. type 2 D. type 3

ugcnetcse-dec2007-paper2

Answer key

40.1 Closure Property (3)

40.1.1 Closure Property: UGC NET CSE | December 2012 | Part 3 | Question: 38

Which is not the correct statement?

A. The class of regular sets is closed under homomorphisms
B. The class of regular sets is not closed under inverse homomorphisms
C. The class of regular sets is closed under quotient
D. The class of regular sets is closed under substitution

ugcnetcse-dec2012-paper3 theory-of-computation closure-property

Answer key ☟

40.1.2 Closure Property: UGC NET CSE | January 2017 | Part 3 | Question: 21

Given the following statements:

i. A class of languages that is closed under union and complementation has to be closed under intersection
ii. A class of languages that is closed under union and intersection has to be closed under complementation

Which of the following options is correct?

A. Both (i) and (ii) are false
B. Both (i) and (ii) are true
C. (i) is true, (ii) is false
D. (i) is false, (ii) is true

ugcnetcse-jan2017-paper3 theory-of-computation regular-language closure-property

Answer key ☟

40.1.3 Closure Property: UGC NET CSE | June 2012 | Part 2 | Question: 40

Consider the following statements:

I. Recursive languages are closed under complementation
II. Recursively enumerable languages are closed under union
III. Recursively enumerable languages are closed under complementation

Which of the above statements are true?

A. I only
B. I and II
C. I and III
D. II and III

theory-of-computation closure-property ugcnetcse-june2012-paper2

Answer key ☟

40.2 Complement (1)

40.2.1 Complement: UGC NET CSE | August 2016 | Part 3 | Question: 23

Let $\Sigma = a, b$ and language $L = aa, bb$. Then, the complement of L is

A. $\{\lambda, a, b, ab, ba\} \cup \{w \in \{a, b\}^* || w| > 3\}$
B. $\{a, b, ab, ba\} \cup \{w \in \{a, b\}^* || w| \geq 3\}$
C. $\{w \in \{a, b\}^* || w| > 3\} \cup \{a, b, ab, ba\}$
D. $\{\lambda, a, b, ab, ba\} \cup \{w \in \{a, b\}^* || w| \geq 3\}$

ugcnetcse-aug2016-paper3 theory-of-computation complement

Answer key ☟

40.3 Context Free Grammar (9)

40.3.1 Context Free Grammar: UGC NET CSE | August 2016 | Part 3 | Question: 57

Let $G = (V, T, S, P)$ be a context-free grammar such that every one of its productions is of the form $A \rightarrow \nu$, with $|\nu| = k > 1$. The derivation tree for any string $W \in L(G)$ has a height such that

A. $h < \frac{(|W|-1)}{k-1}$
B. $\log_k |W| \leq h$
C. $\log_k |W| < h < \frac{(|W|-1)}{k-1}$
D. $\log_k |W| \leq h \leq \frac{(|W|-1)}{k-1}$

ugcnetcse-aug2016-paper3 theory-of-computation context-free-grammar

40.3.2 Context Free Grammar: UGC NET CSE | December 2005 | Part 2 | Question: 4

Which sentence can be generated by $S \to d/bA, A \to d/ccA$:

A. bccddd B. aabccd C. ababccd D. abbbd

ugcnetcse-dec2005-paper2 theory-of-computation context-free-grammar grammar

Answer key

40.3.3 Context Free Grammar: UGC NET CSE | December 2009 | Part 2 | Question: 34

Contex-free Grammar (CFG) can be recognized by

(A) Finite state automata
(B) 2-way linear bounded automata
(C) push down automata
(D) both (B) and (C)

ugcnetcse-dec2009-paper2 theory-of-computation context-free-grammar

Answer key

40.3.4 Context Free Grammar: UGC NET CSE | December 2013 | Part 2 | Question: 27

The context free grammar for the language:
$L = \{a^n b^m \mid n \leq m + 3, n \geq 0, m \geq 0\}$ is

A. $S \to aaa \quad A; A \to aAb \mid B, B \to Bb \mid \lambda$
B. $S \to aaaA \mid \lambda, A \to aAb \mid B, B \to Bb \mid \lambda$
C. $S \to aaaA \mid aaA \mid \lambda, A \to aAb \mid B, B \to Bb \mid \lambda$
D. $S \to aaaA \mid aa \quad A \mid aA \mid \lambda, A \to aAb \mid B, B \to Bb \mid \lambda$

ugcnetcse-dec2013-paper2 theory-of-computation context-free-grammar

Answer key

40.3.5 Context Free Grammar: UGC NET CSE | December 2014 | Part 2 | Question: 35

The following Context-Free Grammar (CFG) :

$S \to aB|bA$

$A \to a|as|bAA$

$B \to b|bs|aBB$

will generate

A. Odd numbers of $a's$ and odd numbers of $b's$
B. Even numbers of $a's$ and even numbers of $b's$
C. Equal numbers of $a's$ and $b's$
D. Different numbers of $a's$ and $b's$

ugcnetcse-dec2014-paper2 theory-of-computation context-free-grammar

Answer key

40.3.6 Context Free Grammar: UGC NET CSE | January 2017 | Part 3 | Question: 22

Let $G = (V, T, S, P)$ be a context-free grammer such that every one of its productions is of the form $A \to v$, with $|v| = K > 1$. The derivation tree for any $W \in L(G)$ has a height h such that

A.

$$\log_K |W| \leq h \leq \log_K \left(\frac{|W|-1}{K-1}\right)$$

B.

$\log_K | W |\leq h \leq \log_K (K | W |)$

C.

$\log_K | W |\leq h \leq K \log_K | W|$

D. $\log_K |W |\leq h \leq \left(\frac{|W|-1}{K-1}\right)$

ugcnetcse-jan2017-paper3 context-free-grammar theory-of-computation

Answer key

40.3.7 Context Free Grammar: UGC NET CSE | June 2012 | Part 3 | Question: 28

Which one of the following is not a Greibach Normal form grammar?

(i)
$S \to a \mid bA \mid aA \mid bB$
$A \to a$
$B \to b$

(ii)
$S \to a \mid aA \mid AB$
$A \to a$
$B \to b$

(iii)
$S \to a \mid A \mid aA$
$A \to a$

A. (i) and (ii)
B. (i) and (iii)
C. (ii) and (iii)
D. (i), (ii) and (iii)

ugcnetcse-june2012-paper3 theory-of-computation context-free-grammar

Answer key

40.3.8 Context Free Grammar: UGC NET CSE | June 2013 | Part 3 | Question: 38

For every context free grammar (G) there exists an algorithm that passes any $w \in L(G)$ in number of steps proportional to

A. $ln \mid w|$
B. $|w|$
C. $|w|^2$
D. $|w|^3$

ugcnetcse-june2013-paper3 theory-of-computation context-free-grammar

Answer key

40.3.9 Context Free Grammar: UGC NET CSE | Junet 2015 | Part 3 | Question: 61

A context free grammar for $L = \{w \mid n_0(w) > n_1(w)\}$ is given by:

A. $S \to 0 \mid 0S \mid 1SS$
B. $S \to 0S \mid 1S \mid 0SS \mid 1SS \mid 0 \mid 1$
C. $S \to 0 \mid 0S \mid 1SS \mid S1S \mid SS1$
D. $S \to 0S \mid 1S \mid 0 \mid 1$

ugcnetcse-june2015-paper3 theory-of-computation context-free-grammar

Answer key

40.4 Context Free Language (8)

40.4.1 Context Free Language: UGC NET CSE | December 2011 | Part 2 | Question: 38

Which one of the following statement is false ?

A. Context-free languages are closed under union.
B. Context-free languages are closed under concatenation.
C. Context-free languages are closed under intersection.
D. Context-free languages are closed under Kleene closure.

ugcnetcse-dec2011-paper2 theory-of-computation context-free-language

Answer key

40.4.2 Context Free Language: UGC NET CSE | December 2018 | Part 2 | Question: 36

Consider R to be any regular language and L_1, L_2 be any two context-free languages. Which of the following is correct?

A. $\overline{L_1}$ is context free
B. $\overline{(L_1 \cup L_2)} - R$ is context free
C. $L_1 \cap L_2$ is context free
D. $L_1 - R$ is context free

ugcnetcse-dec2018-paper2 context-free-language theory-of-computation

Answer key

40.4.3 Context Free Language: UGC NET CSE | January 2017 | Part 3 | Question: 23

Given the following two languages:

$L_1 = \{a^n b^n \mid n \geq 0, \; n \neq 100\}$

$L_2 = \{w \in \{a, b, c\}^* \mid n_a(w) = n_b(w) = n_c(w)\}$

Which of the following options is correct?

A. Both L_1 and L_2 are not context free language
B. Both L_1 and L_2 are context free language
C. L_1 is context free language, L_2 is not context free language
D. L_1 is not context free language, L_2 is context free language

ugcnetcse-jan2017-paper3 theory-of-computation context-free-language

Answer key

40.4.4 Context Free Language: UGC NET CSE | July 2016 | Part 3 | Question: 56

Let $L = \{0^n 1^n | n \geq 0\}$ be a context free language. Which of the following is correct?

A. $\overline{L}$ is context free and L^k is not context free for any $k \geq 1$
B. $\overline{L}$ is not context free and L^k is context free for any $k \geq 1$
C. Both $\overline{L}$ and L^k for any $k \geq 1$ are context free
D. Both $\overline{L}$ and L^k for any $k \geq 1$ are not context free

ugcnetcse-july2016-paper3 context-free-language

Answer key

40.4.5 Context Free Language: UGC NET CSE | June 2012 | Part 3 | Question: 39

The following CFG $S \rightarrow aB \mid bA, A \rightarrow a \mid as \mid bAA, B \rightarrow b \mid bs \mid aBB$ generates strings of terminals that have

A. odd number of a's and odd number of b's
B. even number of a's and even number of b's
C. equal number of a's and b's
D. not equal number of a's and b's

ugcnetcse-june2012-paper3 theory-of-computation context-free-language

Answer key

40.4.6 Context Free Language: UGC NET CSE | June 2014 | Part 2 | Question: 09

The context free grammar for the language $L = \{a^n b^m c^k \mid k = \mid n - m \mid, n \geq 0, m \geq 0, k \geq 0\}$ is

A. $S \to S_1 S_3, S_1 \to aS_1 c \mid S_2 \mid \lambda, S_2 \to aS_2 b \mid \lambda, S_3 \to aS_3 b \mid S_4 \mid \lambda, S_4 \to bS_4 c \mid \lambda$
B. $S \to S_1 S_3, S_1 \to aS_1 S_2 c \mid \lambda, S_2 \to aS_2 b \mid \lambda, S_3 \to aS_3 b \mid S_4 \mid \lambda, S_4 \to bS_4 c \mid \lambda$
C. $S \to S_1 \mid S_2, S_1 \to aS_1 S_2 c \mid \lambda, S_2 \to aS_2 b \mid \lambda, S_3 \to aS_3 b \mid S_4 \mid \lambda, S_4 \to bS_4 c \mid \lambda$
D. $S \to S_1 \mid S_3, S_1 \to aS_1 c \mid S_2 \mid \lambda, S_2 \to aS_2 b \mid \lambda, S_3 \to aS_3 b \mid S_4 \mid \lambda, S_4 \to bS_4 c \mid \lambda$

ugcnetcse-june2014-paper2 theory-of-computation context-free-language

Answer key

40.4.7 Context Free Language: UGC NET CSE | June 2019 | Part 2 | Question: 77

How can the decision algorithm be constructed for deciding whether context-free language L is finite?

i. By constructing redundant CFG G in CNF generating language L
ii. By constructing non-redundant CFG G in CNF generating language L
iii. By constructing non-redundant CFG G in CNF generating language $L - \{\wedge\}$ ($\wedge$ stands for null)

Which of the following is correct?

A. i only B. ii only C. iii only D. None of i, ii and iii

ugcnetcse-june2019-paper2 context-free-language

Answer key

40.4.8 Context Free Language: UGC NET CSE | October 2020 | Part 2 | Question: 29

Which of the following statements is true?

A. The union of two context free languages is context free
B. The intersection of two context free languages is context free
C. The complement of a context free language is context free
D. If a language is context free, it can always be accepted by a deterministic pushdown automaton

ugcnetcse-oct2020-paper2 theory-of-computation context-free-language

Answer key

40.5 Context Sensitive (2)

40.5.1 Context Sensitive: UGC NET CSE | December 2015 | Part 3 | Question: 22

The family of context sensitive languages is _____ under union and ____ under reversal

A. closed, not closed
B. not closed, not closed
C. closed, closed
D. not closed, closed

ugcnetcse-dec2015-paper3 theory-of-computation closure-property context-sensitive

Answer key

40.5.2 Context Sensitive: UGC NET CSE | July 2018 | Part 2 | Question: 37

Context sensitive language can be recognized by a

A. Finite state machine
B. Deterministic finite automata
C. Non-deterministic finite automata
D. Linear bounded automata

ugcnetcse-july2018-paper2 theory-of-computation context-sensitive

Answer key

40.6 Decidability (5)

40.6.1 Decidability: UGC NET CSE | December 2018 | Part 2 | Question: 37

Consider the following problems :

A. Whether a finite state automaton halts on all inputs?
B. Whether a given context free language is regular?
C. Whether a Turing machine computes the product of two numbers?

Which one of the following is correct?

A. Only i and iii are undecidable problems
B. Only ii and iii are undecidable problems
C. Only i and ii are undecidable problems
D. i, ii and iii are undecidable problems

ugcnetcse-dec2018-paper2 decidability theory-of-computation

Answer key

40.6.2 Decidability: UGC NET CSE | June 2019 | Part 2 | Question: 79

Which of the following problems is/are decidable problem(s) (recursively enumerable) on turing machine M?

i. G is a CFG with $L(G) = \phi$
ii. There exist two TMs M_1 and M_2 such that $L(M) \subseteq \{L(M_1) \cup L(M_2)\} =$ language of all TMs.
iii. M is a TM that accepts ω using at most $2^{|\omega|}$ cells of tape

A. i and ii only B. i only C. i, ii and iii D. iii only

ugcnetcse-june2019-paper2 decidability turing-machine

Answer key

40.6.3 Decidability: UGC NET CSE | November 2017 | Part 3 | Question: 22

Which of the following problems is undecidable?

A. To determine if two finite automata are equivalent
B. Membership problem for context free grammar
C. Finiteness problem for finite automata
D. Ambiguity problem for context free grammar

ugcnetcse-nov2017-paper3 theory-of-computation decidability

Answer key

40.6.4 Decidability: UGC NET CSE | October 2020 | Part 2 | Question: 55

Let G_1 and G_2 be arbitrary context free languages and R an arbitrary regular language. Consider the following problems:

i. Is $L(G_1) = L(G_2)$?
ii. Is $L(G_2) \leq L(G_1)$?
iii. Is $L(G_1) = R$?

Which of the problems are undecidable?

Choose the correct answer from the options given below:

A. (i) only B. (ii) only C. (i) and (ii) only D. (i), (ii) and (iii)

ugcnetcse-oct2020-paper2 theory-of-computation decidability

Answer key

40.6.5 Decidability: UGC NET CSE | October 2020 | Part 2 | Question: 87

Given below are two statements:

Statement I: The problem "Is $L_1 \wedge L_2 = \phi$?" is undecidable for context sensitive languages L_1 and L_2

Statement II: The problem "Is $W \in L$?" is decidable for context sensitive language L. (where W is a string).

In the light of the above statements, choose the correct answer from the options given below

A. Both Statement I and Statement II are true
B. Both Statement I and Statement II are false
C. Statement I is correct but Statement II is false
D. Statement I is incorrect but Statement II is true

ugcnetcse-oct2020-paper2 theory-of-computation decidability

Answer key

40.7 Expression (1)

40.7.1 Expression: UGC NET CSE | December 2015 | Part 3 | Question: 26

The context free grammar given by
$S \rightarrow XYX$
$X \rightarrow aX \mid bX \mid \lambda$
$Y \rightarrow bbb$
generates the language which is defined by regular expression:

A. $(a+b)^* bbb$
B. $abbb(a+b)^*$
C. $(a+b)^*(bbb)(a+b)^*$
D. $(a+b)(bbb)(a+b)^*$

theory-of-computation regular-expression finite-automata expression ugcnetcse-dec2015-paper3

Answer key

40.8 Finite Automata (10)

40.8.1 Finite Automata: UGC NET CSE | December 2012 | Part 2 | Question: 44

Let L be set accepted by a non-deterministic finite automaton. The number of states in non-deterministic finite automaton is $|Q|$. The maximum number of states in equivalent finite automaton that accepts L is

A. $|Q|$
B. $2|Q|$
C. $2^{|Q|}-1$
D. $2^{|Q|}$

ugcnetcse-dec2012-paper2 theory-of-computation finite-automata

Answer key

40.8.2 Finite Automata: UGC NET CSE | December 2012 | Part 3 | Question: 47

The minimum number of states of the non-deterministic finite automaton which accepts the language $\{abab^n \mid n \geq 0\} \cup \{aba^n \mid n \geq 0\}$ is

A. 3
B. 4
C. 5
D. 6

ugcnetcse-dec2012-paper3 theory-of-computation finite-automata

Answer key

40.8.3 Finite Automata: UGC NET CSE | December 2018 | Part 2 | Question: 32

Consider the language L given by

$$L = \{2^{nk} \mid k > 0, \text{ and n is non-negative integer number }\}$$

The minimum number of states of finite automaton which accepts the language L is

A. n B. $n+1$ C. $\frac{n(n+1)}{2}$ D. 2^n

ugcnetcse-dec2018-paper2 theory-of-computation finite-automata

Answer key

40.8.4 Finite Automata: UGC NET CSE | July 2018 | Part 2 | Question: 31

Two finite state machines are said to be equivalent if they:

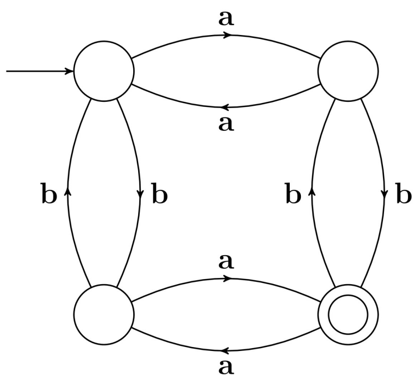
A. Have the same number of edges
B. Have the same number of states
C. Recognize the same set of tokens
D. Have the same number of states and edges

ugcnetcse-july2018-paper2 theory-of-computation finite-automata

Answer key

40.8.5 Finite Automata: UGC NET CSE | July 2018 | Part 2 | Question: 32

The finite state machine given in figure below recognizes:



A. any string of odd number of a's
B. any string of odd number of b's
C. any string of even number of a's and odd number of b's
D. any string of odd number of a's and odd number of b's

ugcnetcse-july2018-paper2 theory-of-computation finite-automata

Answer key

40.8.6 Finite Automata: UGC NET CSE | June 2012 | Part 3 | Question: 16

Given the following statements:

I. The power of deterministic finite state machine and non- deterministic finite state machine are same.
II. The power of deterministic pushdown automaton and non- deterministic pushdown automaton are same.

Which of the above is the correct statement(s)?

A. Both I and II B. Only I C. Only II D. Neither I nor II

ugcnetcse-june2012-paper3 theory-of-computation finite-automata

Answer key

40.8.7 Finite Automata: UGC NET CSE | Junet 2015 | Part 3 | Question: 19

Minimal deterministic finite automaton for the language $L = \{0^n \mid n \geq 0, n \neq 4\}$ will have:

A. 1 final state among 5 states
B. 4 final states among 5 states
C. 1 final state among 6 states
D. 5 final state among 6 states

ugcnetcse-june2015-paper3 theory-of-computation finite-automata

Answer key

40.8.8 Finite Automata: UGC NET CSE | Junet 2015 | Part 3 | Question: 21

The transition function for the language $L = \{w \mid n_a(w) \text{ and } n_b(w) \text{ are both odd}\}$ is given by:

$\delta(q_0, a) = q_1$; $\delta(q_0, b) = q_2$
$\delta(q_1, a) = q_0$; $\delta(q_1, b) = q_3$
$\delta(q_2, a) = q_3$; $\delta(q_2, b) = q_0$
$\delta(q_3, a) = q_2$; $\delta(q_3, b) = q_1$

The initial and final states of the automata are

A. q_0 and q_0 respectively
B. q_0 and q_1 respectively
C. q_0 and q_2 respectively
D. q_0 and q_3 respectively

ugcnetcse-june2015-paper3 theory-of-computation finite-automata

Answer key

40.8.9 Finite Automata: UGC NET CSE | November 2017 | Part 3 | Question: 23

Finite state machine can recognize language generated by _________

A. Only context free grammar
B. Only context sensitive grammar
C. Only regular grammar
D. Any unambiguous grammar

ugcnetcse-nov2017-paper3 theory-of-computation finite-automata

Answer key

40.8.10 Finite Automata: UGC NET CSE | September 2013 | Part 2 | Question: 13

The number of states in a minimal deterministic finite automaton corresponding to the language $L = \{a^n \mid n \geq 4\}$ is

A. 3
B. 4
C. 5
D. 6

ugcnetsep2013ii theory-of-computation finite-automata

Answer key

40.9 Grammar (15)

40.9.1 Grammar: UGC NET CSE | August 2016 | Part 2 | Question: 35

Which of the following is FALSE ?

A. The grammar $S \rightarrow aS|aSbS| \in$, where S is the only non-terminal symbol, and $\in$ is the null string, is ambiguous.
B. An unambiguous grammar has same left most and right most derivation.
C. An ambiguous grammar can never be $LR(k)$ for any k.
D. Recursive descent parser is a top-down parser.

ugcnetcse-aug2016-paper2 theory-of-computation grammar

Answer key

40.9.2 Grammar: UGC NET CSE | December 2012 | Part 3 | Question: 42

The grammar 'GI' $S \rightarrow OSO \mid ISI \mid 0 \mid 1 \mid \in$ and the grammar G2 is $S \rightarrow as \mid asb \mid X, X \rightarrow Xa \mid a$.
Which is the correct statement?

A. G1 is ambiguous, G2 is unambiguous
B. G1 is unambiguous, G2 is ambiguous
C. Both G1 and G2 are ambiguous
D. Both G1 and G2 are unambiguous

ugcnetcse-dec2012-paper3 theory-of-computation grammar

Answer key

40.9.3 Grammar: UGC NET CSE | December 2014 | Part 3 | Question: 62

Match the following :

	List – I		List – II
a.	Context free grammar	i.	Linear bounded automaton
b.	Regular grammar	ii.	Pushdown automaton
c.	Context sensitive grammar	iii.	Turing machine
d.	Unrestricted grammar	iv.	Deterministic finite automaton

Codes :

A. a-ii, b-iv, c-iii, d-i
B. a-ii, b-iv, c-i, d-iii
C. a-iv, b-i, c-ii, d-iii
D. a-i, b-iv, c-iii, d-ii

ugcnetcse-dec2014-paper3 theory-of-computation grammar

Answer key

40.9.4 Grammar: UGC NET CSE | July 2018 | Part 2 | Question: 36

Consider the following two Grammars:

$G_1\ :\ S \rightarrow SbS \mid a$

$G_2 : S \rightarrow aB \mid ab,\ A \rightarrow GAB \mid a,\ B \rightarrow ABb \mid b$

Which one of the folloeing options is correct?

A. Only G_1 is ambiguous
B. Only G_2 is ambiguous
C. Both G_1 and G_2 are ambiguous
D. Both G_1 and G_2 are not ambiguous

ugcnetcse-july2018-paper2 theory-of-computation grammar

Answer key

40.9.5 Grammar: UGC NET CSE | July 2018 | Part 2 | Question: 38

The set $A = \{0^n\ 1^n\ 2^n \mid n = 1, 2, 3, \ldots\}$ is an example of a grammar that is

A. Context sensitive
B. Context free
C. Regular
D. None of the above

ugcnetcse-july2018-paper2 theory-of-computation grammar

Answer key

40.9.6 Grammar: UGC NET CSE | June 2012 | Part 3 | Question: 32

The equivalent grammar corresponding to the grammar $G : S \rightarrow aA, A \rightarrow BB, B \rightarrow aBb \mid \varepsilon$ is

A. $S \rightarrow aA, A \rightarrow BB, B \rightarrow aBb$
B. $S \rightarrow a \mid aA, A \rightarrow BB, B \rightarrow aBb \mid ab$
C. $S \rightarrow a \mid aA, A \rightarrow BB \mid B, B \rightarrow aBb$
D. $S \rightarrow a \mid aA, A \rightarrow BB \mid B, B \rightarrow aBb \mid ab$

theory-of-computation grammar context-free-language ugcnetcse-june2012-paper3

Answer key

40.9.7 Grammar: UGC NET CSE | June 2013 | Part 3 | Question: 35

Which of the following is/are the fundamental semantic model(s) of parameter passing?

A. in mode
B. out mode
C. in-out mode
D. all of the above

ugcnetcse-june2013-paper3 theory-of-computation grammar

Answer key

40.9.8 Grammar: UGC NET CSE | June 2014 | Part 3 | Question: 73

Given the following two grammars :

$G_1 : S \rightarrow AB|aaB$

$A \rightarrow a|Aa$

$B \rightarrow b$

$G_2 : S \rightarrow aSbS|bSaS|\lambda$

Which statement is correct ?

A. G_1 is unambiguous and G_2 is unambiguous.
B. G_1 is unambiguous and G_2 is ambiguous.
C. G_1 is ambiguous and G_2 is unambiguous.
D. G_1 is ambiguous and G_2 is ambiguous.

ugcnetjune2014iii theory-of-computation grammar

Answer key

40.9.9 Grammar: UGC NET CSE | June 2014 | Part 3 | Question: 74

Match the following :

	List – I		List – II
a.	Chomsky Normal form	i.	$S \rightarrow bSS \mid aS \mid c$
b.	Greibach Normal form	ii.	$S \rightarrow aSb \mid ab$
c.	S-grammar	iii.	$S \rightarrow AS \mid a$ $A \rightarrow SA \mid b$
d.	LL grammar	iv.	$S \rightarrow aBSB$ $B \rightarrow b$

Codes :

A. a-iv, b-iii, c-i, d-ii
B. a-iv, b-iii, c-ii, d-i
C. a-iii, b-iv, c-i, d-ii
D. a-iii, b-iv, c-ii, d-i

ugcnetjune2014iii theory-of-computation grammar

Answer key

40.9.10 Grammar: UGC NET CSE | Junet 2015 | Part 2 | Question: 33

If all the production rules have single non-terminal symbol on the left side, the grammar defined is:

A. Context free grammar
B. Context sensitive grammar
C. Unrestricted grammar
D. Phase grammar

ugcnetcse-june2015-paper2 theory-of-computation grammar

Answer key

40.9.11 Grammar: UGC NET CSE | November 2017 | Part 2 | Question: 34

Which of the following statements is/are TRUE?

i. The grammar $S \rightarrow SS \mid a$ is ambiguous. (Where S is the start symbol)
ii. The grammar $S \rightarrow 0S1 \mid 01S \mid \epsilon$ is ambiguous. (The special symbol ϵ represents the empty string) (Where S is the start symbol)
iii. The grammar (Where S is the start symbol)

```
S -> T/U$
T -> x S y | xy | ϵ
U -> yT
```

generates a language consisting of the string $yxxyy$.

A. Only (i) and (ii) are TRUE.
B. Only (i) and (iii) are TRUE.
C. Only (ii) and (iii) are TRUE.
D. All of (i), (ii) and (iii) are TRUE.

ugcnetcse-nov2017-paper2 grammar context-free-language ambiguous

Answer key

40.9.12 Grammar: UGC NET CSE | October 2020 | Part 2 | Question: 57

Which of the following grammars is(are) ambiguous?

i. $s \rightarrow ss \mid asb \mid bsa \mid \lambda$
ii. $s \rightarrow asbs \mid bsas \mid \lambda$
iii.
$s \rightarrow aAB$
$A \rightarrow bBb$
$B \rightarrow A \mid \lambda$ where λ denotes empty string

Choose the correct answer from the options given below:

A. (i) and (iii) only
B. (ii) only
C. (ii) and (iii) only
D. (i) and (ii) only

ugcnetcse-oct2020-paper2 theory-of-computation grammar ambiguous

Answer key

40.9.13 Grammar: UGC NET CSE | September 2013 | Part 2 | Question: 16

LL grammar for the language $L = \{a^n b^m c^{n+m} \mid m \geq 0, n \geq 0\}$ is

A. $S \rightarrow aSc \mid S_1; S_1 \rightarrow bS_1c \mid \lambda$
B. $S \rightarrow aSc \mid S_1\lambda \mid S_1 \mid \lambda; S_1 \rightarrow bS_1c$
C. $S \rightarrow aSc \mid S_1\lambda \mid S_1 \mid \lambda; S_1 \rightarrow bS_1c \mid \lambda$
D. $S \rightarrow aSc \mid S_1\lambda; S_1 \rightarrow bS_1c \mid \lambda$

ugcnetsep2013ii theory-of-computation grammar

Answer key

40.9.14 Grammar: UGC NET CSE | September 2013 | Part 2 | Question: 17

Assume the statements S_1 and S_2 given as:

S_1: Given a context free grammar G, there exists an algorithm for determining whether L(G) is infinite.

S_2: There exists an algorithm to determine whether two context free grammars generate the same language.

Which one of the following is true?

A. S_1 is correct and S_2 is not correct
B. Both S_1 and S_2 are correct
C. Both S_1 and S_2 are not correct
D. S_1 is not correct and S_2 is correct

ugcnetsep2013ii theory-of-computation grammar

Answer key

40.9.15 Grammar: UGC NET CSE | September 2013 | Part 3 | Question: 18

Given the following production of a grammar:

$S \rightarrow aA \mid aBB;$

$A \rightarrow aaA \mid \lambda;$

$B \rightarrow bB \mid bbC;$

$C \rightarrow B$

Which of the following is true?

A. The language corresponding to the given grammar is a set of even number of a's
B. The language corresponding to the given grammar is a set of odd number of a's

C. The language corresponding to the given grammar is a set of even number of a's followed by odd number of b's
D. The language corresponding to the given grammar is a set of odd number of a's followed by even number of b's

ugcnetcse-sep2013-paper3 theory-of-computation grammar

Answer key

40.10 Identify Class Language (8)

40.10.1 Identify Class Language: UGC NET CSE | December 2015 | Part 3 | Question: 28

Given the following two languages:

- $L_1 = \{a^n ba^n \mid n > 0\}$
- $L_2 = \{a^n ba^n b^{n+1} \mid n > 0\}$

Which of the following is correct?

A. L_1 is context free language and L_2 is not context free language
B. L_1 is not context free language and L_2 is context free language
C. Both L_1 and L_2 are context free languages
D. Both L_1 and L_2 are not context free languages

identify-class-language context-free-language ugcnetcse-dec2015-paper3

Answer key

40.10.2 Identify Class Language: UGC NET CSE | December 2019 | Part 2 | Question: 48

Consider the following languages:

$L_1 = \{a^n b^n c^m\} \cup \{a^n b^m c^m\}, n, m \geq 0$

$L_2 = \{ww^R \mid w \in \{a, b\}^*\}$ Where R represents reversible operation.

Which one of the following is (are) inherently ambiguous languages(s)?

A. Only L_1 B. Only L_2 C. both L_1 and L_2 D. neither L_1 nor L_2

ugcnetcse-dec2019-paper2 identify-class-language

Answer key

40.10.3 Identify Class Language: UGC NET CSE | January 2017 | Part 3 | Question: 61

Given the following two statements:

i. $L = \{w \mid n_a(w) = n_b(w)\}$ is deterministic context free language, but not linear.
ii. $L = \{a^n b^n\} \cup \{a^n b^{2n}\}$ is linear, but not deterministic context free language.

Which of the following options is correct?

A. Both (i) and (ii) are false.
B. Both (i) and (ii) are true.
C. (i) is true, (ii) is false.
D. (i) is false, (ii) is true.

ugcnetcse-jan2017-paper3 identify-class-language

Answer key

40.10.4 Identify Class Language: UGC NET CSE | June 2014 | Part 3 | Question: 75

Given the following two languages :

$L_1 = \{a^n b^n | n \geq 1\} \cup \{a\}$

$L_2 = \{wCw^R | w \in \{a, b\}^*\}$

Which statement is correct ?

A. Both L_1 and L_2 are not deterministic.
B. L_1 is not deterministic and L_2 is deterministic.
C. L_1 is deterministic and L_2 is not deterministic.
D. Both L_1 and L_2 are deterministic.

ugcnetjune2014iii theory-of-computation identify-class-language

Answer key

40.10.5 Identify Class Language: UGC NET CSE | October 2020 | Part 2 | Question: 71

Match List I with List II

L_R : Regular language, LCF: Context free language

L_{REC} : Recursive langauge, L_{RE}: Recursively enumerable language.

	List I		List II
(A)	Recursively Enumerable language	(I)	$\overline{L}_{REC} \cup L_{RE}$
(B)	Recursive language	(II)	$\overline{L}_{CF} \cup L_{REC}$
(C)	Context Free language	(III)	$L_R^* \cap L_{CF}$

Choose the correct answer from the options given below:

A. $A-II, B-III, C-I$
B. $A-III, B-I, C-II$
C. $A-I, B-II, C-III$
D. $A-II, B-I, C-III$

ugcnetcse-oct2020-paper2 theory-of-computation identify-class-language

Answer key

40.10.6 Identify Class Language: UGC NET CSE | September 2013 | Part 3 | Question: 19

The language accepted by the non-deterministic pushdown automaton

$M = (\{q_0, q_1, q_2\}, \{a, b\}, \{a, b, z\}, \delta, q_0, z, \{q_2\})$ with transitions

$\delta(q_0 a, z) = \{(q_1 a), (q_2 \lambda)\};$

$\delta(q_1, b, a) = \{(q_1, b)\}$

$\delta(q_1, b, b) = \{(q_1 b)\}, \delta(q_1, a, b) = \{(q_2, \lambda)\}$ is

A. $L(abb * a)$
B. $\{a\} UL(abb * a)$
C. $L(ab * a)$
D. $\{a\} UL(ab * a)$

ugcnetcse-sep2013-paper3 theory-of-computation identify-class-language

Answer key

40.10.7 Identify Class Language: UGC NET CSE | September 2013 | Part 3 | Question: 20

The language $L = \{a^n b^n a^m b^m \mid n \geq 0, m \geq 0\}$ is

A. Context free but not linear
B. Context free and linear
C. Not context free and not linear
D. Not context free but linear

ugcnetcse-sep2013-paper3 theory-of-computation identify-class-language

Answer key

40.10.8 Identify Class Language: UGC NET CSE | September 2013 | Part 3 | Question: 21

Assume statements S_1 and S_2 defined as:

S_1: $L_2 - L_1$ is recursive enumerable where L_1 and L_2 are recursive and recursive enumerable respectively.

S_2: The set of all Turing machines is countable.

Which of the following is true?

A. S_1 is correct and S_2 is not correct
B. Both S_1 and S_2 are correct

C. Both S_1 and S_2 are not correct

D. S_1 is not correct and S_2 is correct

ugcnetcse-sep2013-paper3 identify-class-language theory-of-computation

Answer key

40.11 Language (1)

40.11.1 Language: UGC NET CSE | November 2017 | Part 3 | Question: 24

The language $L = \{a^i\, b\, c^i \mid i \geq 0\}$ over the alphabet $\{a, b, c\}$ is

A. regular language
B. Not a deterministic context free language but a context free language
C. Recursive and is a deterministic context free language
D. Not recursive

ugcnetcse-nov2017-paper3 theory-of-computation language

Answer key

40.12 Languages Classification (1)

40.12.1 Languages Classification: UGC NET CSE | June 2019 | Part 2 | Question: 76

Match List-I with List-II :

where L_1 : Regular language

L_2 : Context-free language

L_3 : Recursive language

L_4 : Recursively enumerable language

	List-I		List-II
(a)	$\overline{L_3} \cup L_4$	(i)	Context-free language
(b)	$\overline{L_2} \cup L_3$	(ii)	Recursively enumerable language
(c)	$L_1^* \cap L_2$	(iii)	Recursive Language

Choose the correct option from those given below :

A. (a)-(ii); (b)-(i); (c)-(iii)

B. (a)-(ii); (b)-(iii); (c)-(i)

C. (a)-(iii); (b)-(i); (c)-(ii)

D. (a)-(i); (b)-(ii); (c)-(iii)

ugcnetcse-june2019-paper2 languages-classification cfl-recusive-re

Answer key

40.13 Languages Relationships (1)

40.13.1 Languages Relationships: UGC NET CSE | December 2014 | Part 3 | Question: 61

Given the recursively enumerable language (L_{RE}), the context sensitive language (L_{CS}), the recursive language (L_{REC}), the context free language (L_{CF}) and deterministic context free language (L_{DCF}). The relationship between these families is given by

A. $L_{CF} \subseteq L_{DCF} \subseteq L_{CS} \subseteq L_{RE} \subseteq L_{REC}$
B. $L_{CF} \subseteq L_{DCF} \subseteq L_{CS} \subseteq L_{REC} \subseteq L_{RE}$
C. $L_{DCF} \subseteq L_{CF} \subseteq L_{CS} \subseteq L_{RE} \subseteq L_{REC}$
D. $L_{DCF} \subseteq L_{CF} \subseteq L_{CS} \subseteq L_{REC} \subseteq L_{RE}$

ugcnetcse-dec2014-paper3 languages-relationships

Answer key

40.14 Minimal State Automata (1)

40.14.1 Minimal State Automata: UGC NET CSE | June 2019 | Part 2 | Question: 75

How many states are there in a minimum state automata equivalent to regular expression given below?
Regular expression is $a^*b(a+b)$

A. 1 B. 2 C. 3 D. 4

ugcnetcse-june2019-paper2 finite-automata minimal-state-automata

Answer key

40.15 Parsing (1)

40.15.1 Parsing: UGC NET CSE | December 2012 | Part 2 | Question: 35

Which of the following is the most powerful parsing method?

A. LL(I) B. Canonical LR C. SLR D. LALR

ugcnetcse-dec2012-paper2 compiler-design parsing

Answer key

40.16 Pumping Lemma (3)

40.16.1 Pumping Lemma: UGC NET CSE | December 2013 | Part 3 | Question: 40

Pumping lemma for context-free languages states:

Let L be an infinite context free language. Then there exists some positive integer m such that any $w \in L$ with $|w| \geq m$ can be decomposed as $w = uvxyZ$ with $|vxy|$ ______ and $|vy|$ ________ such that $uv^{\dot{z}}xy^{\dot{z}}Z \in L$ for all $\dot{z} = 0, 1, 2, \ldots$

A. $\leq m, \leq 1$
B. $\leq m, \geq 1$
C. $\geq m, \leq 1$
D. $\geq m, \geq 1$

ugcnetcse-dec2013-paper3 theory-of-computation context-free-language pumping-lemma

Answer key

40.16.2 Pumping Lemma: UGC NET CSE | November 2017 | Part 3 | Question: 19

The logic of pumping lemma is an example of _________

A. Iteration
B. Recursion
C. The divide and conquer principle
D. The pigeon – hole principle

ugcnetcse-nov2017-paper3 theory-of-computation pumping-lemma

Answer key

40.16.3 Pumping Lemma: UGC NET CSE | November 2017 | Part 3 | Question: 21

Pumping lemma for regular language is generally used for proving

A. Whether two given regular expressions are equivalent
B. A given grammar is ambiguous
C. A given grammar is regular
D. A given grammar is not regular

ugcnetcse-nov2017-paper3 theory-of-computation pumping-lemma

Answer key

40.17 Pushdown Automata (5)

40.17.1 Pushdown Automata: UGC NET CSE | December 2014 | Part 3 | Question: 22

The pushdown automation $M = (\{q_0, q_1, q_2\}, \{a, b\}, \{0, 1\}, \delta, q_0, 0, \{q_0\})$ with

$\delta(q_0, a, 0) = \{(q_1, 10)\}$

$\delta(q_1, a, 1) = \{(q_1, 11)\}$

$\delta(q_1, b, 1) = \{(q_2, \lambda)\}$

$\delta(q_2, b, 1) = \{(q_2, \lambda)\}$

$\delta(q_2, \lambda, 0) = \{(q_0, \lambda)\}$

Accepts the language

A. $L = \{a^n b^m | n, m \geq 0\}$
B. $L = \{a^n b^n | n \geq 0\}$
C. $L = \{a^n b^m | n, m > 0\}$
D. $L = \{a^n b^n | n > 0\}$

ugcnetcse-dec2014-paper3 theory-of-computation pushdown-automata

Answer key

40.17.2 Pushdown Automata: UGC NET CSE | July 2018 | Part 2 | Question: 33

A pushdown automata behaves like a Turing machine when the number of auxiliary memory is

A. 0
B. 1
C. 1 or more
D. 2 or more

ugcnetcse-july2018-paper2 theory-of-computation pushdown-automata

Answer key

40.17.3 Pushdown Automata: UGC NET CSE | July 2018 | Part 2 | Question: 34

Pushdown automata can recognize language generated by _______

A. Only context free grammar
B. Only regular grammar
C. Context free grammar or regular grammar
D. Only context sensitive grammar

ugcnetcse-july2018-paper2 theory-of-computation pushdown-automata

Answer key

40.17.4 Pushdown Automata: UGC NET CSE | June 2013 | Part 3 | Question: 37

A pushdown automation $M = (Q, \Sigma, \Gamma, \delta, q_0, z, F)$ is set to be deterministic subject to which of the following condition(s), for every $q \in Q, a \in \Sigma \cup \{\lambda\}$ and $b \in \Gamma$

(s1) $\delta(q, a, b)$ contains at most one element

(s2) if $\delta(q, \lambda, b)$ is not empty then $\delta(q, c, b)$ must be empty for every $c \in \Sigma$

A. only s1
B. only s2
C. both s1 and s2
D. neither s1 nor s2

ugcnetcse-june2013-paper3 theory-of-computation pushdown-automata

Answer key

40.17.5 Pushdown Automata: UGC NET CSE | September 2013 | Part 3 | Question: 22

Non-deterministic pushdown automaton that accepts the language generated by the grammar: $S \to aSS \mid ab$ is

A. $\delta(q_0, \lambda, z) = \{(q_1, z)\}; \delta(q_0, a, S) = \{(q_1, SS)\}, (q_1, B)\}\delta(q_0, b, B) = \{(q_1, \lambda)\}, \delta(q_1, \lambda, z) = \{(q_f, \lambda)\}$
B. $\delta(q_0, \lambda, z) = \{(q_1, Sz)\}; \delta(q_0, a, S) = \{(q_1, SS)\}, (q_1, B)\}\delta(q_0, b, B) = \{(q_1, \lambda)\}, \delta(q_1, \lambda, z) = \{(q_f, \lambda)\}$
C. $\delta(q_0, \lambda, z) = \{(q_1, Sz)\}; \delta(q_0, a, S) = \{(q_1, S)\}, (q_1, B)\}\delta(q_0, b, \lambda) = \{(q_1, B)\}, \delta(q_1, \lambda, z) = \{(q_f, \lambda)\}$
D. $\delta(q_0, \lambda, z) = \{(q_1, z)\}; \delta(q_0, a, S) = \{(q_1, SS)\}, (q_1, B)\}\delta(q_0, b, \lambda) = \{(q_1, B)\}, \delta(q_1, \lambda, z) = \{(q_f, \lambda)\}$

ugcnetcse-sep2013-paper3 theory-of-computation pushdown-automata

40.18 Recursive and Recursively Enumerable Languages (3)

40.18.1 Recursive and Recursively Enumerable Languages: UGC NET CSE | January 2017 | Part 3 | Question: 63

Which of the following statements is false?

A. Every context-sensitive language is recursive
B. The set of all languages that are not recursively enumerable is countable

C. The family of recursively enumerable language is closed under union
D. The families of recursively enumerable and recursive languages are closed under reversal

ugcnetcse-jan2017-paper3 theory-of-computation recursive-and-recursively-enumerable-languages

Answer key

40.18.2 Recursive and Recursively Enumerable Languages: UGC NET CSE | Junet 2015 | Part 3 | Question: 62

Given the following two statements:

S_1: If L_1 and L_2 are recursively enumerable languages over Σ^*, then $L_1 \cup L_2$ and $L_1 \cap L_2$ are also recursively enumerable.

S_2: The set of recursively enumerable languages is countable.

Which of the following is true?

A. S_1 is correct and S_2 is not correct
B. S_1 is not correct and S_2 is correct
C. Both S_1 and S_2 are not correct
D. Both S_1 and S_2 are correct

ugcnetcse-june2015-paper3 theory-of-computation recursive-and-recursively-enumerable-languages

Answer key

40.18.3 Recursive and Recursively Enumerable Languages: UGC NET CSE | October 2020 | Part 2 | Question: 27

Consider $L = L_1 \cap L_2$ where

$L_1 = \{0^m 1^m 2 0^n 1^n \mid m, n \geq 0\}$

$L_2 = \{0^m 1^n 2^k \mid m, n, k \geq 0\}$

Then, the language L is

A. Recursively enumerable but not context free
B. Regular
C. Context free but not regular
D. Not recursive

ugcnetcse-oct2020-paper2 theory-of-computation recursive-and-recursively-enumerable-languages

Answer key

40.19 Regular Expression (18)

40.19.1 Regular Expression: UGC NET CSE | August 2016 | Part 2 | Question: 31

The number of strings of length 4 that are generated by the regular expression $(0+1+|2+3+)^*$, where | is an alternation character and $+,^*$ are quantification characters, is :

A. 08
B. 09
C. 10
D. 12

ugcnetcse-aug2016-paper2 theory-of-computation regular-expression

Answer key

40.19.2 Regular Expression: UGC NET CSE | August 2016 | Part 3 | Question: 24

Consider the following identities for regular expressions :

(a) (r + s)* = (s + r)*

(b) (r*)* = r*

(c) (r* s*)* = (r + s)*

Which of the above identities are true ?

A. (a) and (b) only
B. (b) and (c) only
C. (c) and (a) only
D. (a), (b) and (c)

ugcnetcse-aug2016-paper3 theory-of-computation regular-expression

Answer key

40.19.3 Regular Expression: UGC NET CSE | December 2005 | Part 2 | Question: 5

Regular expression $a+b$ denotes the set :

A. $\{a\}$
B. $\{\in, a, b\}$
C. $\{a, b\}$
D. None of these

ugcnetcse-dec2005-paper2 theory-of-computation regular-expression

Answer key

40.19.4 Regular Expression: UGC NET CSE | December 2014 | Part 3 | Question: 24

Regular expression for the complement of language $L=\{a^n b^m \mid n \geq 4, m \leq 3\}$ is

A. $(a+b)^* ba(a+b)^*$
B. $a^* bbbbb^*$
C. $(\lambda+a+aa+aaa)b^* + (a+b)^* ba(a+b)^*$
D. None of the above

ugcnetcse-dec2014-paper3 theory-of-computation regular-expression regular-language

Answer key

40.19.5 Regular Expression: UGC NET CSE | January 2017 | Part 2 | Question: 31

Which of the following strings would match the regular expression : $p+[3-5]*[xyz]$?

I. $p443y$
II. $p6y$
III. $3xyz$
IV. $p35z$
V. $p353535x$
VI. $ppp5$

A. I, III and VI only
B. IV, V and VI only
C. II, IV and V only
D. I, IV and V only

ugcnetjan2017ii regular-expression theory-of-computation

Answer key

40.19.6 Regular Expression: UGC NET CSE | July 2016 | Part 2 | Question: 31

The number of strings of length 4 that are generated by the regular expression

$$(0 \mid \epsilon)1^+2^*(3 \mid \epsilon)$$

, where $\mid$ is an alternation character, $\{+, *\}$ are quantification characters, and ϵ is the null string, is:

A. 08
B. 10
C. 11
D. 12

ugcnetcse-july2016-paper2 theory-of-computation regular-expression

Answer key

40.19.7 Regular Expression: UGC NET CSE | July 2016 | Part 3 | Question: 23

The regular expression for the complement of the language $L=\{a^n b^m \mid n \geq 4, m \leq 3\}$ is:

A. $(\lambda+a+aa+aaa)b^* + a^* bbbb^* + (a+b)^* ba(a+b)^*$
B. $(\lambda+a+aa+aaa)b^* + a^* bbbbb^* + (a+b)^* ab(a+b)^*$
C. $(\lambda+a+aa+aaa) + a^* bbbbb^* + (a+b)^* ab(a+b)^*$
D. $(\lambda+a+aa+aaa)b^* + a^* bbbbb^* + (a+b)^* ba(a+b)^*$

ugcnetcse-july2016-paper3 theory-of-computation regular-expression

Answer key

40.19.8 Regular Expression: UGC NET CSE | July 2016 | Part 3 | Question: 24

Consider the following two languages:

$L_1 = \{0^i 1^j \mid gcd(i, j) = 1\}$

L_2 is any subset of 0*

Which of the following is correct?

A. L_1 is regular and L_2* is not regular
B. L_1 is not regular and L_2* is regular
C. Both L_1 and L_2* are regular languages
D. Both L_1 and L_2* are not regular languages

ugcnetcse-july2016-paper3 regular-language regular-expression

Answer key

40.19.9 Regular Expression: UGC NET CSE | July 2016 | Part 3 | Question: 55

Let L be the language generated by regular expression 0*10* and accepted by the deterministic finite automata M. Consider the relation R_M defined by M as all states that are reachable from the start state. R_M has ___ equivalence classes.

A. 2
B. 4
C. 5
D. 6

ugcnetcse-july2016-paper3 theory-of-computation regular-expression regular-language

Answer key

40.19.10 Regular Expression: UGC NET CSE | June 2010 | Part 2 | Question: 3

"My Laughter Machine (MLM) recognizes the following strings :

(i) a

(ii) aba

(iii) $abaabaaba$

(iv) $abaabaabaabaabaabaabaaba$

Using this as an information, how would you compare the following regular expressions ?

(i) $(aba)^{3x}$

(ii) $a.(baa)3^x{-}1.ba$

(iii) $ab.(aab).^{3x-1}\ .a$

A. (ii) and (iii) are same, (i) is different.
B. (ii) and (iii) are not same.
C. $(i), (ii)$ and (iii) are different.
D. $(i), (ii)$ and (iii) are same.

ugcnetcse-june2010-paper2 theory-of-computation regular-expression

Answer key

40.19.11 Regular Expression: UGC NET CSE | June 2012 | Part 3 | Question: 35

Consider the regular expression (a+b)(a+b) ..... (a+b) (n-times). The minimum number of states in finite automaton that recognizes the language represented by this regular expression contains

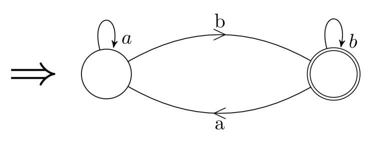
A. n states
B. n+1 states
C. n+2 states
D. 2^n states

ugcnetcse-june2012-paper3 theory-of-computation regular-expression

Answer key

40.19.12 Regular Expression: UGC NET CSE | June 2012 | Part 3 | Question: 43

The regular expression for the following DFA



A. ab*(b+aa*b)*
B. a*b(b+aa*b)*
C. a*b(b*+aa*b)
D. a*b(b+aa*b)*

ugcnetcse-june2012-paper3 theory-of-computation regular-expression

Answer key

40.19.13 Regular Expression: UGC NET CSE | Junet 2015 | Part 3 | Question: 20

The regular expression corresponding to the language L where $L = \{x \in \{0,1\}^* \mid x \text{ ends with 1 and does not contain substring 00 is}$

A. (1+01)* (10+01)
B. (1+01)* 01
C. (1+01)* (1+01)
D. (10+01)* 01

ugcnetcse-june2015-paper3 theory-of-computation regular-expression

Answer key

40.19.14 Regular Expression: UGC NET CSE | November 2017 | Part 2 | Question: 33

Which of the following regular expressions, each describing a language of binary numbers (MSB to LSB) that represents non-negative decimal values, does **not** include even values?

A. $0^*1^+0^*1^*$
B. $0^*1^*0^+1^*$
C. $0^*1^*0^*1^+$
D. $0^+1^*0^*1^*$

Where $\{+, *\}$ are quantification characters.

ugcnetcse-nov2017-paper2 regular-expression regular-language

Answer key

40.19.15 Regular Expression: UGC NET CSE | October 2020 | Part 2 | Question: 28

Let L_1 and L_2 be languages over $\Sigma = \{a, b\}$ represented by the regular expressions $(a^* + b)^*$ and $(a + b)^*$ respectively.

Which of the following is true with respect to the two languages?

A. $L_1 \subset L_2$
B. $L_2 \subset L_1$
C. $L_1 = L_2$
D. $L_1 \cap L_2 = \phi$

ugcnetcse-oct2020-paper2 theory-of-computation regular-expression

Answer key

40.19.16 Regular Expression: UGC NET CSE | October 2020 | Part 2 | Question: 79

Consider the following regular expressions:

i. $r = a(b + a)^*$
ii. $s = a(a + b)^*$
iii. $t = aa^*b$

Choose the correct answer from the options given below based on the relation between the languages generated by the regular expressions above:

A. $L(r) \subseteq L(s) \subseteq L(t)$
B. $L(r) \supseteq L(s) \supseteq L(t)$
C. $L(r) \supseteq L(t) \supseteq L(s)$
D. $L(s) \supseteq L(t) \supseteq L(r)$

ugcnetcse-oct2020-paper2 theory-of-computation regular-expression

Answer key

40.19.17 Regular Expression: UGC NET CSE | September 2013 | Part 2 | Question: 14

Regular expression for the language $L = w \in \{0, 1\}* \mid w \text{ has no pair of consecutive zeros } \}$ is

A. $(1 + 010)*$
B. $(01 + 10)*$
C. $(1 + 010) * (0 + \lambda)$
D. $(1 + 01) * (0 + \lambda)$

ugcnetsep2013ii theory-of-computation regular-expression

Answer key

40.19.18 Regular Expression: UGC NET CSE | September 2013 | Part 2 | Question: 15

Consider the following two languages:

$L_1 = \{a^n b^l a^k \mid n+l+k > 5\}$

$L_2 = \{a^n b^l a^k \mid n > 5, l > 3, k \leq l\}$

Which one of the following is true?

A. L_1 is regular language and L_2 is not regular language
B. Both L_1 and L_2 are regular language
C. Both L_1 and L_2 are not regular language
D. L_1 is not regular language and L_2 is regular language

ugcnetsep2013ii theory-of-computation regular-expression regular-language

Answer key

40.20 Regular Grammar (4)

40.20.1 Regular Grammar: Doubt UGC NET

https://gateoverflow.in/13365/ugcnet-dec2014-iii-24

i've a small doubt in the solution of this question

how is (a+b)*ba(a+b)* complement of the given language?

regular-language regular-grammar query

Answer key

40.20.2 Regular Grammar: UGC NET CSE | August 2016 | Part 3 | Question: 22

The regular grammar for the language $L = \{a^n b^m \mid n+m \text{ is even}\}$ is given by

(1) $S \rightarrow S_1 | S_2$

$S_1 \rightarrow aS_1 | A_1$

$A_1 \rightarrow bA_1 | \lambda$

$S_2 \rightarrow aaS_2 | A_2$

$A_2 \rightarrow bA_2 | \lambda$

(2) $S \rightarrow S_1 | S_2$

$S_1 \rightarrow aS_1 | aA_1$

$S_2 \rightarrow aaS_2 | A_2$

$A_1 \rightarrow bA_1 | \lambda$

$A_2 \rightarrow bA_2 | \lambda$

(3) $S \rightarrow S_1 | S_2$

$S_1 \rightarrow aaaS_1 | aA_1$

$S_2 \rightarrow aaS_2 | A_2$

$A_1 \to bA_1|\lambda$

$A_2 \to bA_2|\lambda$

(4) $S \to S_1|S_2$

$S_1 \to aaS_1|A_1$

$S_2 \to aaS_2|aA_2$

$A_1 \to bbA_1|\lambda$

$A_2 \to bbA_2|b$

ugcnetcse-aug2016-paper3 theory-of-computation regular-grammar

Answer key

40.20.3 Regular Grammar: UGC NET CSE | June 2014 | Part 2 | Question: 10

The regular grammar for the language L= { $w \mid n_a$(w) and $n_b(w)$ are both even, $w \in \{a,b\}$ * } is given by : (Assume, p, q, r and s are states)

A. $p \to aq \mid br \mid \lambda, q \to bs \mid apr \to as \mid bp, s \to ar \mid bq$, p and s are initial and final states.
B. $p \to aq \mid br, q \to bs \mid apr \to as \mid bp, s \to ar \mid bq$, p and s are initial and final states.
C. $p \to aq \mid br \mid \lambda, q \to bs \mid apr \to as \mid bp, s \to ar \mid bq$ p is both initial and final states.
D. $p \to aq \mid br, q \to bs \mid apr \to as \mid bp, s \to ar \mid bq$ p is both initial and final states.

ugcnetcse-june2014-paper2 theory-of-computation regular-grammar

Answer key

40.20.4 Regular Grammar: UGC NET CSE | September 2013 | Part 3 | Question: 17

A regular grammar for the language $L = \{a^n b^m \mid \text{ n is even and m is even}\}$ is given by

A. $S \to aSb \mid S_1; S_1 \to bS_1a \mid \lambda$
B. $S \to aaS \mid S_1; S_1 \to bSb \mid \lambda$
C. $S \to aSb \mid S_1; S_1 \to S_1ab \mid \lambda$
D. $S \to aaS \mid S_1; S_1 \to bbS_1 \mid \lambda$

ugcnetcse-sep2013-paper3 theory-of-computation regular-grammar

Answer key

40.21 Regular Language (9)

40.21.1 Regular Language: UGC NET CSE | August 2016 | Part 3 | Question: 55

Given the following two languages :

$L_1 = uww^R\nu|u, v, w \in (a,b)^+$

$L_2 = uww^R\nu|u, \nu, w \in (a,b)^+, |u| \geq |\nu|$

Which of the following is correct ?

A. L_1 is regular language and L_2 is not regular language.
B. L_1 is not regular language and L_2 is regular language.
C. Both L_1 and L_2 are regular languages.
D. Both L_1 and L_2 are not regular languages.

ugcnetcse-aug2016-paper3 theory-of-computation regular-language

Answer key

40.21.2 Regular Language: UGC NET CSE | December 2013 | Part 2 | Question: 28

Given the following statements:

S_1: If L is a regular language then the language $\{uv \mid u \in L, v \in L^R\}$ is also regular.

S_2: $L = \{ww^R\}$ is regular language.

Which of the following is true?

A. S_1 is not correct and S_2 is not correct
B. S_1 is not correct and S_2 is correct
C. S_1 is correct and S_2 is not correct
D. S_1 is correct and S_2 is correct

ugcnetcse-dec2013-paper2 theory-of-computation regular-language

Answer key

40.21.3 Regular Language: UGC NET CSE | December 2014 | Part 3 | Question: 23

Given two languages :

$L_1 = \{(ab)^n a^k | n > k, k \geq 0\}$

$L_2 = \{a^n b^m | n \neq m\}$

Using pumping lemma for regular language, it can be shown that

A. L_1 is regular and L_2 is not regular.
B. L_1 is not regular and L_2 is regular.
C. L_1 is regular and L_2 is regular.
D. L_1 is not regular and L_2 is not regular.

ugcnetcse-dec2014-paper3 theory-of-computation regular-language

Answer key

40.21.4 Regular Language: UGC NET CSE | January 2017 | Part 3 | Question: 19

Which of the following are not regular?

i. Strings of even number of a's
ii. Strings of a's , whose length is a prime number.
iii. Set of all palindromes made up of a's and b's.
iv. Strings of a's whose length is a perfect square.

A. (i) and (ii) only
B. (i), (ii) and (iii) only
C. (ii),(iii) and (iv) only
D. (ii) and (iv) only

ugcnetcse-jan2017-paper3 theory-of-computation regular-language

Answer key

40.21.5 Regular Language: UGC NET CSE | January 2017 | Part 3 | Question: 20

Consider the languages $L_1 = \phi$ and $L_2 = \{1\}$. Which one of the following represents $L_1^* \cup L_2^* L_1^*$?

A. $\{\in\}$
B. $\{\in, 1\}$
C. ϕ
D. 1^*

ugcnetcse-jan2017-paper3 theory-of-computation regular-language

Answer key

40.21.6 Regular Language: UGC NET CSE | July 2016 | Part 3 | Question: 22

The symmetric differences of two sets S_1 and S_2 is defined as:

$S_1 \oplus S_2 = \{x \mid x \in S_1 \text{ or } x \in S_2, \text{ but x is not in both } S_1 \text{ and } S_2\}$

The nor of two languages is defined as

$nor(L_1, L_2) = \{w \mid w \notin L_1 \text{ and } w \notin L_2\}$

Which of the following is correct?

A. The family of regular languages is closed under symmetric difference but not closed under nor
B. The family of regular languages is closed under nor but not closed under symmetric difference
C. The family of regular languages are closed under both symmetric difference and nor
D. The family of regular languages are not closed under both symmetric difference and nor

ugcnetcse-july2016-paper3 theory-of-computation regular-language

Answer key

40.21.7 Regular Language: UGC NET CSE | June 2014 | Part 3 | Question: 15

Let L be any language. Define $\text{Even}(W)$ as the strings obtained by extracting from W the letters in the even-numbered positions and $\text{Even}(L) = \{\text{Even}(W) \mid W \in L\}$. We define another language $\text{Chop}(L)$ by removing the two leftmost symbols of every string in L given by $\text{Chop}(L) = \{W \mid vW \in L, \text{ with } |v| = 2\}$. If L is regular language then

A. Even(L) is regular and Chop(L) is not regular
B. Both Even(L) and Chop(L) are regular
C. Even(L) is not regular and Chop(L) is regular
D. Both Even(L) and Chop(L) are not regular

ugcnetjune2014iii theory-of-computation regular-language

Answer key

40.21.8 Regular Language: UGC NET CSE | October 2020 | Part 2 | Question: 56

Consider the following languages:

$L_1 = \{a^{\grave{z}^z} \mid \grave{Z} \text{ is an integer}\}$

$L_2 = \{a^{z\grave{z}} \mid \grave{Z} \geq 0\}$

$L_3 = \{\omega\omega \mid \omega\epsilon\{a, b\}^*\}$

Which of the languages is(are) regular?

Choose the correct answer from the options given below:

A. L_1 and L_2 only
B. L_1 and L_3 only
C. L_1 only
D. L_2 only

ugcnetcse-oct2020-paper2 theory-of-computation regular-language

Answer key

40.21.9 Regular Language: UGC NET CSE | September 2013 | Part 3 | Question: 16

Assume, L is regular language. Let statements S_1 and S_2 be defined as:

$S_1 : SQRT(L) = \{x \mid \text{ for some y with } \mid y \mid = \mid x^2 \mid, xy \in L\}$ is regular.

$S_2 : LOG(L) = \{x \mid \text{ for some y with } \mid y \mid = 2^{|x|}, xy \in L\}$ is regular.

Which of the following is true?

A. S_1 is correct and S_2 is not correct
B. Both S_1 and S_2 are correct
C. Both S_1 and S_2 are not correct
D. S_1 is not correct and S_2 is correct

ugcnetcse-sep2013-paper3 theory-of-computation regular-language

Answer key

40.22 Right Quotient (1)

40.22.1 Right Quotient: UGC NET CSE | June 2013 | Part 2 | Question: 20

Given L1=L(a*baa*) L2=L(ab*). The regular expression corresponding to language L3=L1/L2 (right quotient) is given by

A. a*b
B. a*baa*
C. a*ba*
D. None of the above

ugcnetcse-june2013-paper2 theory-of-computation right-quotient

Answer key

40.23 Strings (1)

40.23.1 Strings: UGC NET CSE | June 2019 | Part 2 | Question: 78

Consider the following grammar:

$S \rightarrow XY$

$X \rightarrow YaY \mid a$ and $Y \rightarrow bbX$

Which of the following statements is/are true about the above grammar?

i. Strings produced by the grammar can have consecutive three a's.
ii. Every string produced by the grammar have alternate a and b.
iii. Every string produced by the grammar have at least two a's.
iv. Every string produced by the grammar have b's in multiple of 2.

A. i only B. ii and iii only C. iv only D. iii and iv only

ugcnetcse-june2019-paper2 grammar strings

Answer key

40.24 Turing Machine (5)

40.24.1 Turing Machine: UGC NET CSE | August 2016 | Part 3 | Question: 56

Given a Turing Machine

$M = (q_0, q_1, 0, 1, 0, 1, B, \delta, B, q_1)$

Where δ is a transition function defined as

$\delta(q_0, 0) = (q_0, 0, R)$

$\delta(q_0, B) = (q_1, B, R)$

The language $L(M)$ accepted by Turing machine is given as :

A. 0*1* B. 00* C. 10* D. 1*0*

ugcnetcse-aug2016-paper3 theory-of-computation turing-machine

Answer key

40.24.2 Turing Machine: UGC NET CSE | January 2017 | Part 3 | Question: 62

Which of the following pairs have different expressive power?

A. Single-tape-turing machine and multi-dimensional turing machine
B. Multi-tape-turing machine and multi-dimensional turing machine
C. Deterministic push down automata and non-deterministic push down automata
D. Deterministic finite automata and non-deterministic finite automata

ugcnetcse-jan2017-paper3 theory-of-computation turing-machine

Answer key

40.24.3 Turing Machine: UGC NET CSE | July 2016 | Part 3 | Question: 57

Given a Turing Machine

$M = (\{q_0, q_1, q_2, q_3\}, \{a, b\}, \{a, b, B\}, \delta, B, \{q_3\})$

where δ is a atransaction function defined as

$\delta(q_0, a) = (q_1, a, R)$

$\delta(q_1, b) = (q_2, b, R)$

$\delta(q_2, a) = (q_2, a, R)$
$\delta(q_2, b) = (q_3, b, R)$
The language L(M) accepted by the Turing machine is given as:

A. aa*b
B. abab
C. aba*b
D. aba*

ugcnetcse-july2016-paper3 theory-of-computation turing-machine

Answer key

40.24.4 Turing Machine: UGC NET CSE | July 2018 | Part 2 | Question: 40

Consider the following statements( ):

S_1: There exist no algorithm for deciding if any two Turing machine M_1 and M_2 accept the same language

S_2: The problem of determining whether a Turing machine halts on any input is undecidable

Which one of the following options is correct?

A. Both S_1 and S_2 are correct
B. Both S_1 and S_2 are not correct
C. Only S_1 is correct
D. Only S_2 is correct

ugcnetcse-july2018-paper2 theory-of-computation turing-machine

Answer key

40.24.5 Turing Machine: UGC NET CSE | June 2019 | Part 2 | Question: 80

For a statement

A language $L \subseteq \Sigma^*$ is recursive if there exists some turing machine M. Which of the following conditions is satisfied for any string ω?

A. If $\omega \in L$, then M accepts ω and M will not halt
B. If $\omega \notin L$, then M accepts ω and M will halt by reaching at final state
C. If $\omega \notin L$, then M halts without reaching to acceptable state
D. If $\omega \in L$, then M halts without reaching to an acceptable state

ugcnetcse-june2019-paper2 turing-machine

Answer key

Answer Keys

40.0.1	D	40.0.2	A	40.0.3	B	40.0.4	C	40.0.5	B
40.0.6	B	40.0.7	B	40.0.8	B	40.0.9	C	40.0.10	A
40.0.11	C	40.0.12	B	40.0.13	A	40.0.14	C	40.0.15	C
40.0.16	C	40.0.17	B	40.0.18	B	40.0.19	D	40.0.20	Q-Q
40.0.21	Q-Q	40.0.22	A	40.0.23	Q-Q	40.0.24	Q-Q	40.0.25	Q-Q
40.0.26	A	40.0.27	Q-Q	40.1.1	B	40.1.2	C	40.1.3	B
40.2.1	D	40.3.1	Q-Q	40.3.2	Q-Q	40.3.3	Q-Q	40.3.4	D
40.3.5	X	40.3.6	D	40.3.7	C	40.3.8	D	40.3.9	C
40.4.1	Q-Q	40.4.2	Q-Q	40.4.3	C	40.4.4	C	40.4.5	C
40.4.6	D	40.4.7	C	40.4.8	A	40.5.1	B	40.5.2	Q-Q
40.6.1	Q-Q	40.6.2	C	40.6.3	D	40.6.4	D	40.6.5	A
40.7.1	B	40.8.1	D	40.8.2	C	40.8.3	Q-Q	40.8.4	Q-Q
40.8.5	Q-Q	40.8.6	B	40.8.7	D	40.8.8	D	40.8.9	C

40.8.10	C
40.9.5	Q-Q
40.9.10	A
40.9.15	B
40.10.5	C
40.12.1	B
40.16.2	D
40.17.4	C
40.19.1	Q-Q
40.19.6	D
40.19.11	B
40.19.16	B
40.20.3	C
40.21.4	C
40.21.9	B
40.24.3	C

40.9.1	Q-Q
40.9.6	D
40.9.11	D
40.10.1	X
40.10.6	B
40.13.1	D
40.16.3	D
40.17.5	B
40.19.2	Q-Q
40.19.7	D
40.19.12	D
40.19.17	D
40.20.4	D
40.21.5	D
40.22.1	C
40.24.4	Q-Q

40.9.2	B
40.9.7	D
40.9.12	D
40.10.2	Q-Q
40.10.7	A
40.14.1	C
40.17.1	B
40.18.1	B
40.19.3	Q-Q
40.19.8	B
40.19.13	C
40.19.18	A
40.21.1	Q-Q
40.21.6	C
40.23.1	D
40.24.5	C

40.9.3	B
40.9.8	D
40.9.13	C
40.10.3	B
40.10.8	B
40.15.1	B
40.17.2	Q-Q
40.18.2	D
40.19.4	D
40.19.9	D
40.19.14	C
40.20.1	Q-Q
40.21.2	C
40.21.7	B
40.24.1	Q-Q

40.9.4	Q-Q
40.9.9	C
40.9.14	A
40.10.4	D
40.11.1	C
40.16.1	B
40.17.3	Q-Q
40.18.3	C
40.19.5	D
40.19.10	Q-Q
40.19.15	C
40.20.2	Q-Q
40.21.3	D
40.21.8	X
40.24.2	C